

OpenADMET-PXR Challenge

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Members:

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Track 1: Activity Prediction

We worked with a filtered train set of 3390 compounds. For the final models, the Phase-1 unblinded set was included in our train set. We used consistent 90:10 scaffold split throughout. Docking with Autodock was performed for all train and test instances and docking scores were obtained. The following models were trained:

1. GNN model:

Atom and Bond features follow our earlier methods (chemrxiv: <https://doi.org/10.26434/chemrxiv-2025-1tb2m>), and trained with early stopping on the validation dataset (~120 epochs). Predictions on the 10% validation set (376 compounds) and test set of 513 (incl. 260 unblinded) compounds were obtained.

2. Chemprop:

Used the available Chemprop implementation to train on both pEC50 and Emax, since this dual task might enable better learning on the shared parameters. Predictions on the 10% validation set (376 compounds) and test set of 513 (incl. 260 unblinded) compounds were obtained.

3. TabPFN model with mordred-derived features:

We computed the full set of 1826 mordred descriptors for all compounds in the train and test sets, including all structural descriptors. We optimized for feature completeness for all train instances and obtained ~1500 features. The missing values in these features were imputed using an iterative randomforest imputer with 3 iterations (on the train set only, without the targets). The same imputers were applied on the testset for its missing values. This completed dataset was augmented with docking scores and then used for learning with TabPFN. Predictions on the 10% validation set (376 compounds) and test set of 513 (incl. 260 unblinded) compounds were obtained.

4. XGB models:

Two XGB models were trained that differed in their data :

Model 1: The completed dataset of mordred features described above, along with docking scores

Model 2: A composite of mordred + morgan + maccs, along with docking scores.

For both models, hyperparameters for both models were optimized using 10-fold CV MAE gridsearch with early-stopping on a hyperdimensional voluminous grid. For both models, predictions on the 10% validation set (376 compounds) and test set of 513 (incl. 260 unblinded) compounds were obtained.

Ensemble model. Using the predictions on the validation set of 376 compounds, an L1 Convex Blend ensembler was optimized, yielding weights for each model. It is noted that only one of the two XGB models turned out significant, so four different models / model classes were used in the final ensemble. The optimized model weights were then applied to the test predictions from each model to yield the final submitted test predictions.

Track 2: Structure Prediction

We used PXR monomer (LBD) as well as PXR dimer (LBD+DBD dimer) to cofold with the ligand using Boltz, Chai, and Protenix. Based on iPTM and pLDDT, Boltz was consistently better (on all complexes), so we narrowed to analyzing Boltz-cofolded monomer and dimer PXR with ligand complexes. From the dimer complexes, we extracted the bound monomer + ligand structure. We then compared the binding site and pose in the pure monomer complex and the extracted monomer complex, to identify consensus and divergence. Where there was consensus, we used the pure monomer complex, and in the case of divergence, we settled for the extracted monomer complex. This was examined by posebusters and violations in non-aromatic ring non-flatness were observed (absence of necessary pucker). The final structures were minimized and fixed for bond orders.

Scripts for the models and model performances are noted in additional resources and will be updated.

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